

WEN JIN LEE

US Citizen | 📞 (678) 882-4269 | ✉️ jinlee2@andrew.cmu.edu | [in linkedin.com/in/wenjinlee](https://www.linkedin.com/in/wenjinlee) | 🌐 [My website](#)

Education

Carnegie Mellon University

B.A.Sc. Chemical Engineering

Expected: May 2028

Pittsburgh, PA

Work Experience

Gold Peak Technology Group Ltd. (GP EnergyTech)

Jun 2026 – Aug 2026

Manufacturing & Cell Engineering Intern

Hong Kong & China

- Fabricated Nickel-Zinc coin cells end-to-end — electrode coating, cell assembly, and electrochemical characterization via galvanostatic cycling and **Electrochemical Impedance Spectroscopy (EIS)**
- Performed failure analysis on Ni-Zn cells to diagnose key degradation mechanisms, including zinc dendrite formation, hydrogen evolution reaction (HER), electrode passivation, and active material dissolution
- Assigned to a team of 5, responsible for supervising and directing a factory with **500 workers**, creating SOPs, engineering assembly lines, and maintaining supply chain for NiMH, Li-ion, and Ni-Zn batteries
- Designed assembly line configuration for newly innovated Ni-Zn battery packs using **AutoCAD**, and fabricated chassis for 16-384 cell battery packs using **Fusion360**, assembled with 3D printing and soldering

CMU Robotics Institute - Soft Machines Lab (Upcoming)

Starting Oct 2026

Undergraduate Researcher

Pittsburgh, PA

- Will conduct research on gecko-inspired padding for soft aquatic robots in Dr. Carmel Majidi's Soft Machines Lab (SML)

Projects

Perfumery

Mar 2024 – Present

- Managing a social media platform with **7K+ followers, 50K+ likes, and 1.5M+ views**, creating educational content on technical fragrance analysis and reviews of commercial perfumes
- Handling and studying over **200** aromachemicals, natural oils, and synthetic chemicals, documenting olfactory notes, IFRA restrictions, CAS numbers, and chemical budgeting in Excel
- Analyzed fragrance samples using **GC-MS** results, interpreting chromatograms and mass spectra to identify constituent chemicals and understand adverse effects of chemical reactions on the perfume
- **Current status:** Founding an independent brand and accepted as a contract perfumer for Xyrena Fragrances

Propulsion Engineering | Carnegie Mellon Rocket Command

Aug 2026 – Present

- Managing fluid system design for bipropellant ethanol/nitrous oxide Half-Cat style liquid rocket engine

Solar Cell Capture | Carnegie Mellon Solar Racing

Aug 2026 – Present

- Optimizing solar cell energy capture for use in a competitive solar-powered boat

Venturi Flow Meter | Fluid Mechanics (06-261)

Apr 2026 – May 2026

- Worked in a team of 4, fully designed a Venturi flow meter in **Fusion360** and 3D printed using 1500 strong resin
- Modeled flow meter to model blood flow in an aortic aneurysm and validated results against **COMSOL** simulations

Going Bananas! | Lab 1 (09-221) Team Project

Mar 2026 – May 2026

- Implemented **atomic absorption spectroscopy (AAS)** to assess potassium precipitate composition for fertilizer use

L'Oréal Brandstorm

Jan 2026 – Mar 2026

- Designed skeleton for a bionic wrist device using **Fusion360** that implements piezo disks to dispense fragrances
- Created formulas with compounds to relieve depressive symptoms, citing research on modulating GABA-A receptors

Leadership

President of CMU Singapore Student Association (SSA)

Apr 2026 – Present

- Leading a club with **500+** current & past students, and managing a fund of **\$20,000+** per year for cultural events
- Acting as primary liaison between SSA, the Singapore Government, HSBC, and RedDotLink

Orientation Counselor (OC) & Transfer OC

Apr 2026 – Sep 2026

- Orientation counselor for incoming CMU class, 1 of 2 transfer OCs for specially tailored transfer orientation of 64 students

Tartan Cultural League (TCL) Representative

Apr 2026 – Present

- Sole representative for SSA in a multicultural league responsible for coordinating tournaments and activities for students

Technical Skills and Languages

Languages: English (native), Chinese (conversational), Bahasa Melayu (basic)

Skills: MATLAB, Fusion360, Python, COMSOL, AutoCAD, AAS, HPLC, GC, LaTeX, Aspen Plus, EIS, MS Office Suite